

RRT* Motion Planning with Control Barrier Function Safety Derived from Poisson Fields

Maciej Rózewicz

Abstract—This paper proposes a novel framework for safe motion planning that integrates Control Barrier Function (CBF) safety constraints with the Rapidly-Exploring Random Tree Star (RRT*) algorithm. The CBF is derived directly from the solution of a Poisson equation defined over the free workspace, referred to as a *Poisson Safety Function (PSF)*. The PSF encodes obstacle boundaries and safety gradients in a smooth, differentiable field that naturally satisfies Laplace-type spatial regularity. The proposed method, termed *Safe-CBF-RRT**, uses the PSF and its spatial derivatives to enforce pCBF-based safety conditions along sampled edges during tree expansion. Unlike pure safety-filter approaches such as FaSTrack, our method actively generates an inherently safe path while maintaining computational tractability. Simulation results demonstrate that the method enhances safety and convergence properties compared to standard RRT*.

I. INTRODUCTION

Ensuring safe navigation in complex environments is a major challenge in modern robotics, particularly in scenarios involving human-robot interaction and autonomous vehicles. Traditional motion planning algorithms often focus on finding feasible paths without formal guarantees of safety. To achieve such provable safety, Control Barrier Functions (CBFs) have emerged as a powerful tool for enforcing safety constraints in control systems [1].

In parallel, methods like FaSTrack [4] provide formal guarantees for safe trajectory tracking by defining a Safe-to-Track Set (S). However, these are primarily safety filters for the execution phase and do not inherently provide an efficient planner to generate the safe reference trajectory itself, often suffering from high computational cost in high-dimensional state spaces.

Sampling-based motion planners such as RRT* [6] are widely used for complex robotic systems due to their scalability. However, they rely primarily on geometric collision checking, offering no formal guarantees of safety. Integrating CBFs into sampling-based planners is a promising direction to enhance safety while retaining the benefits of these methods.

A critical challenge in using CBFs is the construction of a suitable, continuously differentiable safety function $h(x)$. We address this by generating $h(x)$ as the solution of a Poisson equation [3], yielding a Poisson Safety Function (PSF) that encodes obstacle boundaries and provides smooth safety gradients. This definition of $h(x)$ is advantageous for complex environments where analytical forms are not feasible and is naturally compatible with the RRT* framework.

Contributions: The *Safe-CBF-RRT** framework makes the following contributions:

- Integration of the PSF into a modified RRT* algorithm to guide safety-aware path expansion.
- A computationally efficient alternative to tracking-based safety methods (like FaSTrack) for generating inherently safe reference trajectories.
- Empirical results showing improved safety margins and convergence properties over standard RRT*.

II. RELATED WORK

Sampling-based methods like RRT* [6] are standard in high-dimensional planning but lack formal safety guarantees. Potential field and PDE-based navigation [7] ensure path smoothness but do not provide a safety margin measurable in the CBF framework [1]. Our approach bridges this gap by generating the continuously differentiable safety function $h(x)$ through a Poisson PDE [2], [3] that respects complex boundary geometry.

A. Integration of CBFs in Sampling-based Planning

CBF integration in planning typically falls into two categories:

- **Control Synthesis (Dynamic Methods):** Approaches like Robust CBF-RRT* [8] solve Quadratic Programming (QP) problems at each step to find a dynamically safe control input u . While guaranteeing safety for kinodynamic systems, this incurs a high online computational cost during tree expansion.
- **Path Filtering (Geometric Methods):** Our method, *Safe-CBF-RRT**, uses a computationally lighter, simplified, discrete CBF condition applied to the path segment geometry. This shifts the focus from finding an optimally safe control to efficiently generating an inherently safe reference path.

This approach contrasts with the FaSTrack framework [4], which uses Hamilton-Jacobi-Isaacs (HJI) equations to compute a Safe-to-Track Set (S). FaSTrack decouples planning from safety assurance, requiring an expensive feasibility check after the path is generated. By integrating the Poisson-derived safety constraint directly into the sampling and rewiring steps, *Safe-CBF-RRT** efficiently generates inherently safe paths, avoiding the need for expensive high-dimensional HJI solutions characteristic of FaSTrack-type approaches.

III. PRELIMINARIES

A. RRT* Algorithm

The RRT* algorithm [6] is an incremental, sampling-based motion planner that asymptotically converges to the optimal feasible path. Unlike RRT, RRT* guarantees optimality through two key operations: (1) Optimal Parent Selection: choosing the parent x_{parent} for the new node x_{new} from neighbors within radius r_n that minimizes the total cost-to-come; and (2) Rewiring: checking if neighbors can reduce their cost by adopting x_{new} as a new parent. The connection radius $r_n = \gamma(\frac{\log n}{n})^{\frac{1}{d}}$ ensures probabilistic completeness and convergence to the optimal cost.

B. Control Barrier Functions (CBFs)

Control Barrier Functions (CBFs) [5] provide a formalism for ensuring safety in control systems by enforcing forward invariance of a safe set. Given a dynamical system

$$\dot{x} = f(x) + g(x)u, \quad (1)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the control input, and f, g are locally Lipschitz continuous functions, a set

$$\mathcal{C} = \{x \in \mathbb{R}^n : h(x) \geq 0\} \quad (2)$$

is defined as *safe* if there exists a continuously differentiable function $h : \mathbb{R}^n \rightarrow \mathbb{R}$ such that for all $x \in \mathcal{C}$,

$$\sup_{u \in U} [L_f h(x) + L_g h(x)u] \geq -\kappa h(x), \quad (3)$$

where $L_f h$ and $L_g h$ are the Lie derivatives of h along f and g , respectively, and $\kappa > 0$ is a tuning parameter - the smaller κ is the more conservative is condition. This condition ensures that the system's trajectories remain within the safe set $\mathcal{C}$ for all future times.

C. Poisson Safety Function (PSF) Generation

Let $\Omega \subset \mathbb{R}^2$ denote the workspace with boundary $\partial\Omega$ and obstacles $\mathcal{O}_i$. We define a vector potential field $\mathbf{u} = [u_x, u_y]^T$ solving

$$\left\{ \begin{array}{ll} \nabla^2 \mathbf{u} = 0 & \text{in } \Omega \\ \mathbf{u} = c_i(\mathbf{x} - \mathbf{c}_i) & \text{on } \partial\mathcal{O}_i \end{array} \right\}. \quad (4)$$

A scalar field $h(x, y)$, termed the *Poisson Safety Function (PSF)*, is then obtained by solving

$$\left\{ \begin{array}{ll} \nabla^2 h = -\|\nabla \mathbf{u}\| & \text{in } \Omega \\ h|_{\partial\Omega} = 0 & \text{on } \partial\mathcal{O}_i \end{array} \right\}, \quad (5)$$

ensuring $h > 0$ in free space. Gradients $\nabla h = [\partial h / \partial x, \partial h / \partial y]$ are computed using finite differences.

IV. SAFE CBF-RRT* ALGORITHM

We introduce several key modifications to the classical RRT* algorithm in order to incorporate safety constraints derived from the Poisson-based Control Barrier Function.

The most important changes are:

- 1) **Edges are validated using a CBF condition** computed from the Poisson Safety Function (PSF) and its spatial gradients.

- 2) **A constant nominal traversal speed is assumed** when moving along each candidate edge. This assumption is important: slower motion leads to larger feasible sets under the CBF condition, meaning that *even narrow passages may be safe at lower velocities*, while the same passages are considered unsafe at higher speeds.
- 3) **Edges violating the CBF constraint are rejected**, preventing unsafe tree expansion even if the geometric collision check passes.
- 4) **A safety-weighted cost** is introduced to prefer paths that stay in regions with higher PSF values (more safety margin).

Together, these changes turn the sampling process itself into a safety-aware procedure, rather than merely filtering an already planned path.

A. Ensuring Safety with Poisson CBFs

The Poisson Safety Function $h(x)$ and its spatial derivatives provide a smooth, differentiable representation of obstacle boundaries and safety gradients. In Safe CBF-RRT*, these quantities are used in three ways:

- **Discrete CBF validation of each candidate edge.** For a short edge between nodes a and b , we assume motion with constant velocity $\dot{x} = v(b - a)/|b - a|$. The CBF condition —, sampled at multiple points p along the edge, determines whether traversal is safe. This directly links feasibility to assumed speed.
- **Safety-aware cost weighting.** Each edge is assigned a cost depending on both its length and average safety margin $\bar{h}$.
- **Optional control-feasibility check.** For edges that satisfy discrete CBF conditions, PSF gradients can be used to verify whether a continuous control input exists to maintain safety along the smooth trajectory.

B. Trajectory Cost with Safety Weighting

To encourage the planner to use safer portions of the workspace, Safe CBF-RRT* modifies the classical RRT* cost function as:

$$J(a, b) = cL + (1 - c)\frac{1}{\bar{h}}, \quad (6)$$

where $L = |b - a|$ is the geometric length, and $\bar{h}$ is the average PSF value along the edge. Parameter $c \in [0, 1]$ determines the trade-off between path efficiency and safety.

C. Summary of Safe-CBF-RRT*

The overall algorithm follows the standard RRT* structure but rejects edges that fail the CBF feasibility condition (dependent on assumed traversal speed). This results in a tree that grows preferentially through regions of high safety margin while still preserving the asymptotic optimality guarantees of RRT* for the modified cost.

V. RESULTS

This section presents simulation results comparing Safe-CBF-RRT* against standard RRT* in a 2D environment with randomly located obstacles. It provides diversity of paths, safety margins, and path lengths.

A. Simulation Setup

The workspace was a square randomly located obstacles. The grid resolution was 100×100 . Parameters used were $\text{stepSize} = 2$, $\kappa = 20$, and $N_{\text{iter}} = 2000$.

There were three scenarios with varying obstacle densities and configurations:

- **Scenario I:** Moderately spaced obstacles with few wide corridors.
- **Scenario II:** Tightly packed obstacles with narrow passages and one wide corridor.
- **Scenario III:** Few bigger obstacles with one good, wide corridor.

Each scenario was run 100 times for both algorithms - it means with $c = 1$ (length optimized only) and $c = 0$ (safety optimized only), recording path length, average safety metric $\bar{h}$, number of iterations, and rejection ratio to show how many potentially unsafe segments are rejected from tree. Scenarios with $c = 1$ are run with two velocities: $2 \frac{m}{s}$ and $0.5 \frac{m}{s}$ to show ability of Safe-CBF-RRT* to find safer paths when moving slower.

Additionally for both algorithm calibrations corresponding iterations were run with same random seed to show how both algorithms behave in exactly same conditions.

For simulation purposes, a simple unicycle kinematic model was assumed:

$$\begin{cases} \dot{x} = v \cos(\theta) \\ \dot{y} = v \sin(\theta) \\ \dot{\theta} = \omega \end{cases}. \quad (7)$$

B. Performance

1) *Scenario I:* Scenario I represents an environment with moderately spaced obstacles and relatively wide free-space corridors. The PSF field (Fig. 1) exhibits smooth gradients and well-defined safety basins around obstacles, making it a representative baseline configuration. Standard RRT* tends

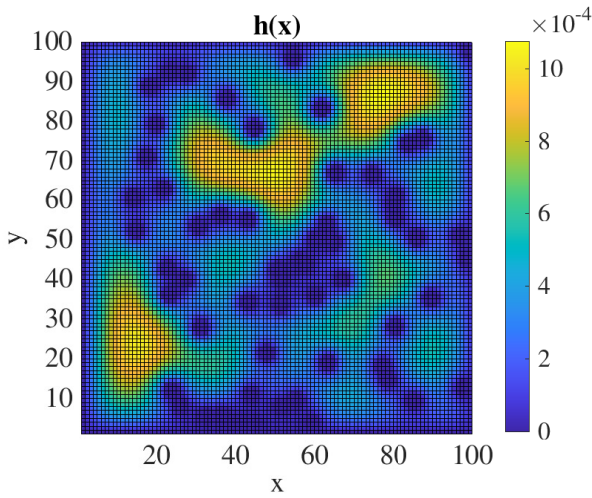


Fig. 1. Control barrier function for Scenario I.

to exploit narrow gaps, often producing shorter but riskier paths (Fig. 2). This behavior is reflected in the low average

safety value ($\bar{h} = 0.00057$). In contrast, Safe-CBF-RRT* systematically rejects unsafe edges (48.7% of samples), forcing the tree to expand through regions of higher PSF values (Fig. 4). As a result, the mean path length increases from 118.57 to 147.86, while the mean safety metric improves by 16%. The higher number of iterations used by CBF-

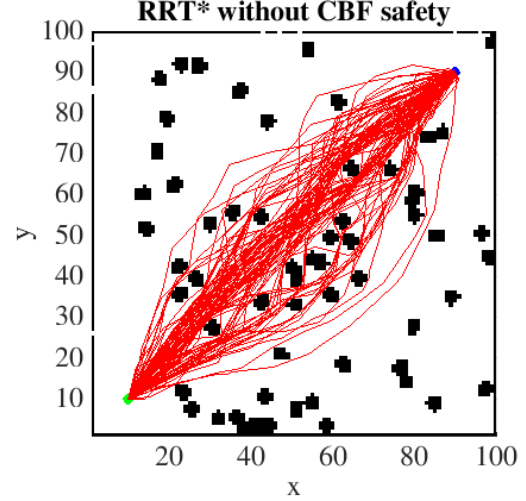


Fig. 2. Exemplary results of 100 different runs of standard RRT* algorithm for Scenario I.

RRT* is consistent with the stronger filtering mechanism, which prioritizes safety over direct progress toward the goal. Overall, Scenario I demonstrates that the proposed method effectively increases safety margins even in relatively easy environments. For the reduced target speed of $0.5 \frac{m}{s}$, the

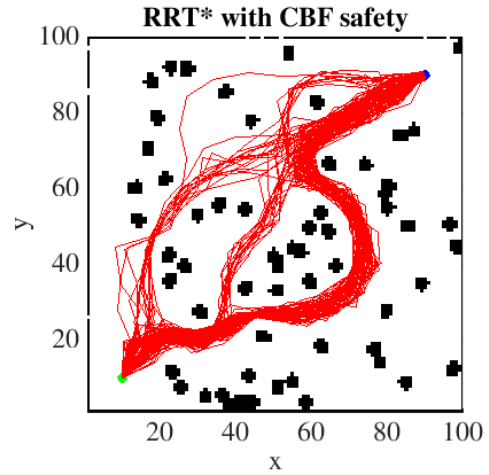


Fig. 3. Exemplary results of 100 different runs of CBF-RRT* algorithm for Scenario I - for $2 \frac{m}{s}$ target speed.

impact of the CBF constraint becomes noticeably weaker. The rejection ratio decreases from 48.7% to 27.5%, and the required number of iterations drops accordingly (372). The resulting paths remain significantly safer than those generated by standard RRT* ($\bar{h} = 0.00065$), but their geometric

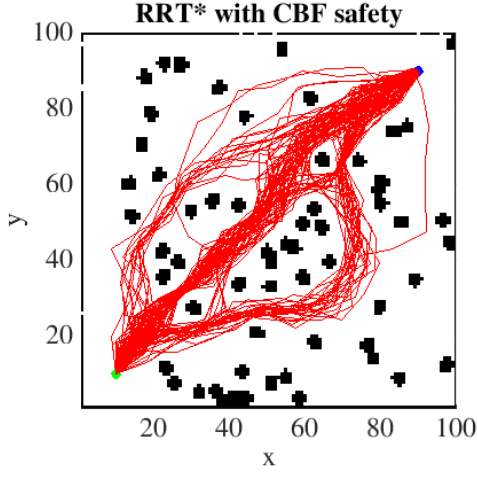


Fig. 4. Exemplary results of 100 different runs of CBF-RRT* algorithm for Scenario I - for $0.5 \frac{m}{s}$ target speed.

length increases only moderately (128.49), remaining much closer to the baseline RRT* solution. This indicates that at lower target speeds the environment effectively becomes less restrictive, allowing Safe-CBF-RRT* to satisfy the safety constraints without forcing large detours. Consequently, the planner converges more efficiently while still benefiting from the PSF-based safety shaping.

TABLE I
PERFORMANCE COMPARISON IN SCENARIO I.

Algorithm	Length	$\bar{h}(x)$	Iterations	Rejections
RRT*	118.57	0.00057	333	13.6%
CBF-RRT* ($2 \frac{m}{s}$)	147.86	0.00066	563	48.7%
CBF-RRT* ($0.5 \frac{m}{s}$)	128.49	0.00065	372	27.5%

2) *Scenario II*: Scenario II introduces significantly tighter passages and higher obstacle density. The PSF field (Fig. 5) shows steeper gradients and narrow channels of admissible motion, making safety enforcement more restrictive. Standard RRT* frequently selects unsafe extensions in this environment, reflected by a much higher rejection ratio (29.0%) compared to Scenario I. Its solutions remain relatively short (119.84), but operate close to obstacles, maintaining only a moderate safety average ($\bar{h} = 0.00085$). Safe-CBF-RRT* must navigate through a more constrained safety landscape, leading to an even larger rejection ratio (59.9%) and correspondingly more conservative motion. Although the resulting paths are longer (130.95), they traverse regions with noticeably higher PSF values, improving the average safety metric by 29% relative to standard RRT*. Scenario II highlights the ability of the method to maintain safe operation under geometric constraints that naturally encourage risk-taking behavior in classical sampling-based planners. When the target speed is reduced to $0.5 \frac{m}{s}$, the CBF filtering remains strong but becomes considerably less restrictive. The rejection ratio drops from 59.9% to 48.5%, and the number of iterations decreases from 537 to 407, indicating that the

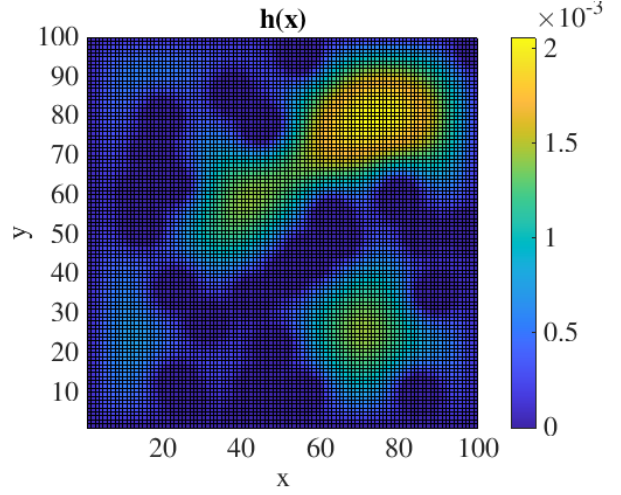


Fig. 5. Control barrier function for Scenario II.

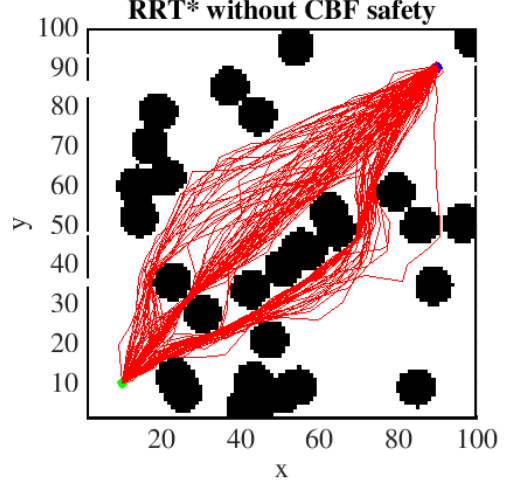


Fig. 6. Exemplary 100 different runs of standard RRT* algorithm for Scenario II.

planner requires fewer alternative expansions to locate a safe homotopy class. The resulting trajectories exhibit nearly identical length (130.80) and safety level ($\bar{h} = 0.00109$) as in the $2 \frac{m}{s}$ case. This shows that even in highly constrained environments, lower forward velocity relaxes the safety constraint sufficiently to reduce computational effort while preserving the safety benefits of the proposed method.

3) *Scenario III*: Scenario III consists primarily of an open environment with a few concentrated high-risk areas. The PSF distribution (Fig. 9) features broad low-gradient regions interspersed with sharp drops near obstacles. In this configuration, standard RRT* is able to find relatively direct paths (122.71), but due to the scattered obstacle layout, it often crosses regions of low safety potential. This is reflected in the lowest average safety value among all scenarios ($\bar{h} = 0.00037$). Safe-CBF-RRT* improves the safety metric to $\bar{h} = 0.00048$ by rejecting 58.4% of unsafe expansions and steering the tree around low-PSF zones (Fig. 12). In-

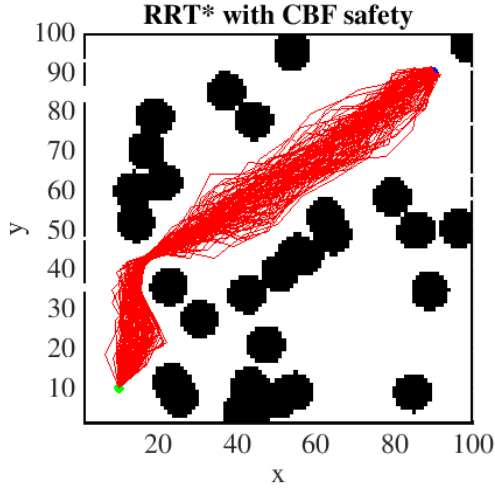


Fig. 7. Exemplary 100 different runs of CBF-RRT* algorithm for Scenario II - for $2 \frac{m}{s}$ target speed.

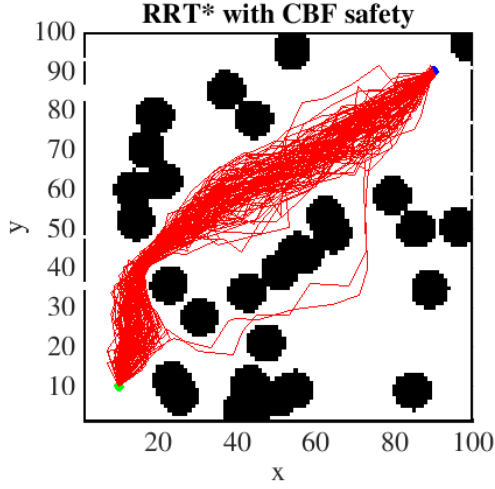


Fig. 8. Exemplary 100 different runs of CBF-RRT* algorithm for Scenario II - for $0.5 \frac{m}{s}$ target speed.

terestingly, the number of iterations is slightly lower than for standard RRT*, indicating that large open areas reduce the need for rewiring and costly exploration. Despite the safety constraints, the method only marginally increases the path length (from 122.71 to 126.32), demonstrating that when safe alternatives are abundant, the penalty for enforcing CBF constraints becomes minimal. Scenario III shows that the proposed planner adapts gracefully to environments where safe path options are naturally available, avoiding unnecessary conservatism while still improving safety. For the reduced speed of $0.5 \frac{m}{s}$, the planner benefits from the abundance of safe alternatives: the rejection ratio decreases from 58.4% to 48.7%, and the number of iterations further drops to 362. Despite this, the safety level remains essentially unchanged ($\bar{h} = 0.00047$), and the resulting trajectory length (126.51) stays very close to the $2 \frac{m}{s}$ variant. This confirms that in open environments the influence of the CBF constraint

TABLE II
PERFORMANCE COMPARISON IN SCENARIO II.

Algorithm	Length	$h(x)$	Iterations	Rejections
RRT*	119.84	0.00085	345	29.0%
CBF-RRT* ($2 \frac{m}{s}$)	130.95	0.00110	537	59.9%
CBF-RRT* ($0.5 \frac{m}{s}$)	130.80	0.00109	407	48.5%

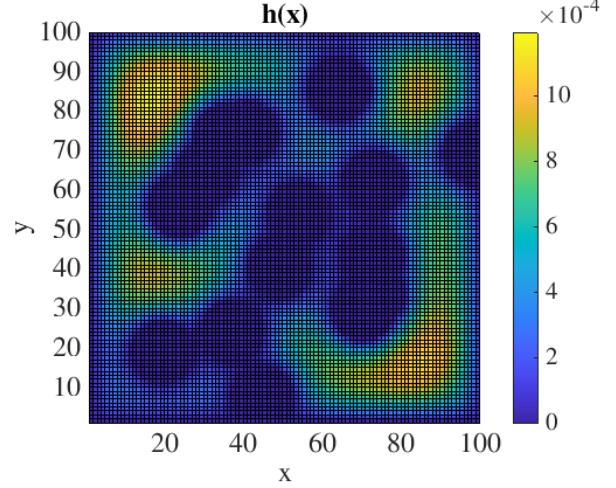


Fig. 9. Exemplary 100 different runs of standard RRT* algorithm.

diminishes at lower speeds, since safe feasible expansions are readily available, and the planner does not require significant detours to maintain conservative operation.

TABLE III
PERFORMANCE COMPARISON IN SCENARIO III.

Algorithm	Length	$h(x)$	Iterations	Rejections
RRT*	122.71	0.00037	421	39.8%
CBF-RRT* ($2 \frac{m}{s}$)	126.32	0.00048	389	58.4%
CBF-RRT* ($0.5 \frac{m}{s}$)	126.51	0.00047	362	48.7%

C. Discussion

The experimental evaluation across the three scenarios demonstrates that integrating a Poisson-based CBF into the RRT* framework substantially reshapes the planner's behavior, consistently favoring safer regions of the workspace. The results reveal a clear trade-off between geometric path length and improved safety ($\bar{h}$).

In Scenario I (moderately constrained), the high rejection ratio (nearly 50%) confirms that the PSF actively filters out geometrically efficient but unsafe edges, ensuring consistently safer trajectories at the cost of moderate path elongation. The algorithm maintains stable convergence despite increased filtering.

In Scenario II (highly constrained), standard RRT* forces paths through narrow, unsafe gaps. In contrast, Safe-CBF-RRT* actively prunes unsafe topological options early on (highest rejection ratio), leading the planner to favor longer detours through alternative, smoother PSF corridors. This

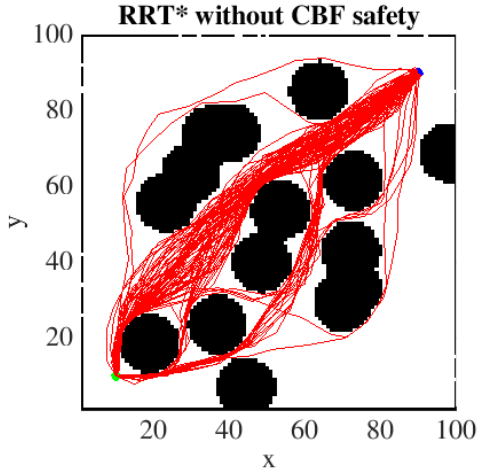


Fig. 10. Exemplary 100 different runs of standard RRT* algorithm for Scenario III.

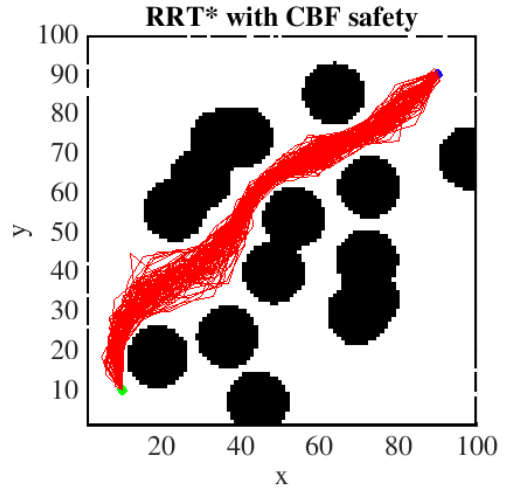


Fig. 12. Exemplary 100 different runs of CBF-RRT* algorithm for Scenario III - for $0.5 \frac{m}{s}$ target speed.

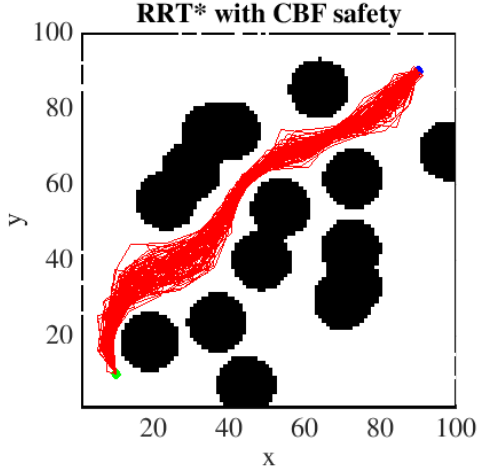


Fig. 11. Exemplary 100 different runs of CBF-RRT* algorithm for Scenario III - for $2 \frac{m}{s}$ target speed.

highlights the benefit of embedding CBF reasoning directly in the sampling process, which forces exploration into safer homotopy classes. Consequently, this scenario exhibits the largest safety improvement and highest variance in path length.

In Scenario III (open environment), the difference between planners is minimal, as most edges are already safely admissible. The rejection ratio drops, and path lengths converge. Nevertheless, the safety-weighted cost successfully biases the rewiring step toward more robust trajectories, providing a marginal safety improvement at virtually no additional planning cost.

Across all experiments, the higher mean number of iterations for Safe-CBF-RRT* confirms the method influences the global structure of the growing tree rather than acting as a lightweight post-filter. Finally, the parameter c allows continuous tuning between pure length optimization ($c = 1$) and pure risk minimization ($c = 0$), where the latter

strongly prefers wide, smooth-value PSF basins. Overall, the results confirm that Poisson-based CBF integration provides a principled mechanism for steering the sampling process toward inherently safe trajectories, increasing robustness in complex environments while preserving the asymptotic optimality structure of RRT*.

VI. CONCLUSION AND FUTURE WORK

We presented *Safe-CBF-RRT**, a motion planning framework combining RRT* with a Control Barrier Function derived from a Poisson Safety Field. The approach unifies PDE-based potential field generation with formal safety guarantees **in a manner computationally lighter than tracking-based methods (e.g., FaSTrack)**, while actively shaping the path for safety. Future work will extend this method to kinodynamic models, uncertain environments, and real robotic experiments.

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